Training and Model-Selection Details

Supplementary record for `main.tex`

10 September 2026

1 Configuration summary

System	Input and representation	Training configuration	Validation selection
Full TF–IDF	Complete user turns; word 1–2-grams (150k) and character 3–5-grams (200k)	Class-balanced logistic regression; selected $C = 1$ from $\{0.25, 1, 4\}$	Guidance threshold 0.51 selected on validation data
Full ModernBERT	Complete user turns; base and large encoders; maximum length 4,096	Batch size 32, weight decay 0.01, seed 42; guidance: 3 epochs at 5×10^{-5}; class-weighted topic: 10 epochs at 6×10^{-5}	Checkpoint selected by macro-F1; guidance thresholds: 0.26 (base), 0.38 (large)
Prefix TF–IDF	User-turn prefixes; word 1–2-grams and character-within-word 3–5-grams	Logistic regression with $C = 0.25$; conversation-level weighting	Weighted macro-F1; threshold 0.42
Prefix ModernBERT	User-turn prefixes; base and large encoders; maximum length 4,096	Batch size 32, weight decay 0.01, seed 42; 3 epochs at 5×10^{-5}; conversation-level weighting	Weighted validation loss; thresholds: 0.51 (base), 0.36 (large)

Table 1: Training configurations. Thresholds apply to guidance detection. Prefix thresholds maximize conversation-weighted macro-F1, using positive-class F1 as a tie-breaker.

The fixed 200-conversation gold evaluation set was sampled with seed 42 using earlier labels to contain 100 guidance-positive and 100 guidance-negative conversations. Topic was not used for this selection. After Sol annotation and human review, it contains 84 guidance-positive and 116 guidance-negative conversations. The remaining 1,922 conversations form the silver development set and were split with seed 42 into 1,632 training and 290 validation conversations. This split was stratified first by guidance and then, among guidance-positive conversations where counts allowed, by primary topic.

2 Prefix construction

A conversation prefix contains all user messages up to a given turn. If guidance begins at user turn k, prefixes before k are negative and prefixes at or after k are positive. Conversations without guidance contribute only negative prefixes. All prefixes from one conversation remain in the same split.

Each prefix is weighted by the inverse of its conversation's number of user turns, so every conversation contributes a total weight of 1. Prefix training is used only for guidance detection; no prefix topic classifier is trained.

The frozen splits contain 4,865 training prefixes from 1,632 conversations and 912 validation prefixes from 290 conversations. Of these, 2,042 training prefixes and 376 validation prefixes are positive.

3 Model selection and hardware

The prefix ModernBERT models were trained on an A100 40GB GPU. Their best checkpoints minimize conversation-weighted validation loss. Guidance thresholds maximize conversation-weighted macro-F1, with positive-class F1 used as a tie-breaker.

Full-conversation ModernBERT checkpoints are selected by validation macro-F1 before guidance-threshold selection. Topic training uses class weights because the legal-topic distribution is imbalanced.

4 Silver annotation provenance

Taxonomy development used OpenAI's GPT-5.6 Terra through the OpenAI API at high reasoning effort. Final silver annotation used GPT-5.6 Sol at medium reasoning effort under the frozen taxonomy. Both runs used the structured prompt and JSON schema in `classification/prompts.py`. The corresponding run configurations are `classification/config_v21_review_balanced_nonguidance_terra.yaml` and `classification/config_v3_classify_all.yaml`.

5 Archival reproducibility record

- Annotation source revision: `837f1001802bcf08ad18baf61ea52f7abfc35b84`.
- Hierarchical training split fingerprints: train `6ba4c2696e889276` and validation `403fe118d76c8360`.
- Gold evaluation: 200 human-reviewed conversations, including 84 guidance-positive conversations.
- Bootstrap: 2,000 conversation-level resamples with replacement, seed 42, without stratification.
- Full TF–IDF: $C = 1$ and guidance threshold 0.51, selected after searching $C \in \{0.25, 1, 4\}$.

6 Human-review completeness

Both reviewers examined all 200 gold conversations. Every case in which their judgments differed was jointly re-examined and resolved by consensus, producing a final guidance label and primary topic where applicable. All 200 conversations are retained in the classifier evaluations.

7 Artifact collection and evaluation code

The Legal Guidance in the Wild Hugging Face collection consolidates the final datasets, checkpoints, training details, and evaluation artifacts. The public evaluation script is `scripts/evaluate_gold.py`. From the repository root, install the dependencies with `pipinstall-rrequirements.txt` and run `pythonscripts/evaluate_gold.py` to recompute the gold-set results from the frozen predictions in `supplementary/evaluation/`.